CLASS 9 · ICSE · SELINA CONCISE MATHEMATICS · FORMULA BANK

Chapter 27: Graphical Solution

Every formula and result of the chapter on one sheet

1	Solution	Point of intersection of the two lines
2	One solution	$\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$
3	No solution	$\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$
4	Infinitely many solutions	$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$
5	Meets the x-axis	Put $y = 0$
6	Meets the y-axis	Put $x = 0$
7	Triangle with the x-axis	$\frac{1}{2} \times$ base on the x-axis $\times \lvert y \rvert$ of the intersection
8	Triangle with the y-axis	$\frac{1}{2} \times$ base on the y-axis $\times \lvert x \rvert$ of the intersection
9	Vertical line	$x = a$
10	Horizontal line	$y = b$
11	Check	Substitute the solution in both equations
12	Number of points per line	Three (two to draw, one to check)